

Problem 1.

1. True. If one set $c=0$, then FALSE, because the network collapses to a constant map, but this is edge case not intended in standard NN.
2. False. Although the NTK limit is taken in the width $\rightarrow \infty$ regime, the kernel itself does depend on depth. The eigen-spectrum of the NTK typically goes "Full Rank" as depth increases, s.t. can fit more complex function.
3. False. Adam can settle at higher loss than well-tuned SGD, it depends on hyper-parameters of optimisation.

Problem 2.

1. trainable weight matrices $W_Q, W_K, W_V \in \mathbb{R}^{8 \times d_h}$, d_h denotes head size.
projection. $Q = XW_Q$, $K = XW_K$, $V = XW_V$
 $Q, K, V \in \mathbb{R}^{4 \times d_h}$. With h heads we replicate the same step for each head and concatenate along last dim.
2. During next-token prediction, the model shall not "peak" future tokens
so for each position t , the weights for keys $> t$ are masked to $-\infty$ (0)
before the softmax. For 4 tokens the causal mask is $M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$
3. Encoder-only (e.g. BERT): The bidirectional self-attention lets every token attend to both left and right, giving stronger representations for classification or span-prediction
4. Decoder-only (e.g. GPT): The causal structure enables straight forward auto-regressive text generation without needing an additional output head.

Problem 3.

1. $W_1 \in \mathbb{R}^{k \times m}$, $b_1 \in \mathbb{R}^{k \times 1}$; $Z = \sigma(W_1 X + b_1) \in \mathbb{R}^{k \times n}$; $W_2 \in \mathbb{R}^{1 \times k}$, $b_2 \in \mathbb{R}$; $y = \sigma(W_2 Z + b_2) \in \mathbb{R}^{1 \times n}$.
2. $\#W_1 = km$, $\#b_1 = k$, $\#W_2 = k$, $\#b_2 = 1 \Rightarrow$ total number = $k(m+2) + 1$
3. common choices: logistic sigmoid $\sigma(u) = \frac{1}{1+e^{-u}}$, ReLU, tanh.
4. (average over n samples) $L = -\frac{1}{n} \sum_{i=1}^n [y^0 i \log y_i + (1-y^0 i) \log(1-y_i)]$, y^0 denote GT. y_i denote $\hat{y}_i$.
5. let $\delta^{(2)} = y - y^0 \in \mathbb{R}^{1 \times n}$ (since σ' cancels for logistic CE). Backprop: $\nabla_{W_2} L = \frac{1}{n} \delta^{(2)} Z^T$, $\nabla_{b_2} L = \frac{1}{n} \sum_i \delta_i^{(2)}$; Hidden layer err. $\delta^{(1)} = (W_2^T \delta^{(2)}) \odot \sigma'(U)$, $U = W_1 X + b_1$.
 $\nabla_{W_1} L = \frac{1}{n} \delta^{(1)} X^T$, $\nabla_{b_1} L = \frac{1}{n} \sum_i \delta_{:,i}^{(1)}$

Problem 4.

1. $1+1+4+10+6+2 = 24$ convolution layers.
2. A max-pool window of 2×2 pixels with stride 2; (no padding)
3. Input $448 \times 448 \times 3$, First conv: 7×7 kernel, stride 2, 64 filters, (padding 3)
 $\frac{448+2 \times 3-7}{2} + 1 = 224$. Hence $224 \times 224 \times 64$.
4. Flattened size = $7 \times 7 \times 1024 = 50176$; $FC_1 = 50176 \times 4096 + 4096 = 205524992$.
 $FC_2 = 4096 \times 1470 + 1470 = 6022590$; Total = $FC_1 + FC_2 = 211547582$.
5. For each of the 7×7 grid cell YOLO predicts $B=2$ bounding boxes. each with 5 numbers (x, y, w, h , confidence) plus $C=20$ class probabilities.
 $B \times 5 + C = 2 \times 5 + 20 = 30$. Thus $7 \times 7 \times 30$ for all predictions.

Problem 5.

1. Write $m = n_s + n_t$, let $\phi(x) \in \mathbb{R}^{m \times \dim \mathcal{H}}$ stack embeddings. Define L as statement.

$$\text{MMD}^2 = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(x_i^s) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi(x_j^t) \right\|_{\mathcal{H}}^2$$

$$\text{MMD}^2(x_s, x_t) = \left\| \mu_s - \mu_t \right\|_{\mathcal{H}}^2 = \langle \mu_s - \mu_t, \mu_s - \mu_t \rangle_{\mathcal{H}}. \text{ Expanding:}$$

$$\langle \mu_s, \mu_s \rangle_{\mathcal{H}} = \frac{1}{n_s^2} \sum_{i=1}^{n_s} \sum_{j=1}^{n_s} \langle \phi(x_i^s), \phi(x_j^s) \rangle_{\mathcal{H}} = \frac{1}{n_s^2} \sum_{i,j} k(x_i^s, x_j^s); \quad \langle \mu_t, \mu_t \rangle_{\mathcal{H}} = \frac{1}{n_t^2} \sum_{i,j} k(x_i^t, x_j^t); \quad \langle \mu_s, \mu_t \rangle_{\mathcal{H}} = \frac{1}{n_s n_t} \sum_{i,j} k(x_i^s, x_j^t)$$

$$\text{Hence } \text{MMD}^2(x_s, x_t) = \frac{1}{n_s^2} \sum_{i,j} k(x_i^s, x_j^s) + \frac{1}{n_t^2} \sum_{i,j} k(x_i^t, x_j^t) - \frac{2}{n_s n_t} \sum_{i,j} k(x_i^s, x_j^t) \quad (\star)$$

Stack the $n_s + n_t$ in order $(x_1, \dots, x_{n_s+n_t})$, X as stated. $L_{pq} = \begin{cases} \frac{1}{n_s^2} & p, q \leq n_s \\ \frac{1}{n_t^2} & p, q > n_s \\ -\frac{1}{n_s n_t} & \text{otherwise} \end{cases}$

Then each double sum in $(\star)$ is a block-wise Frobenius inner product $K_{uv} = L_{u,v}^{-1}$ otherwise

$$K \text{ is symmetric. } \text{MMD}^2(x_s, x_t) = \sum_{p,q} K_{pq} L_{pq} = \langle K, L \rangle_F = \text{tr}(K^T L) = \text{tr}(KL). \quad \text{Q.E.D.}$$

Problem 5.

2. For any $f, g \in \mathcal{H}$ the Cauchy-Schwarz inequality gives $|\langle g, f \rangle| \leq \|g\| \|f\|$. sup gives $g, \|f\|$:
 $\sup_{\|g\| \leq 1} \langle g, f \rangle \leq \sup_{\|g\| \leq 1} \|g\| \|f\| = \|f\|$, equal by $g = f/\|f\|$ (or unit-vector multiple of f)

$$\|f\|_{\mathcal{H}} = \sup_{\|g\|_{\mathcal{H}} \leq 1} \langle g, f \rangle \quad (1)$$

let $\mu_P = \mathbb{E}_{x \sim P} [\phi(x)]$ and $\mu_Q = \mathbb{E}_{x \sim Q} [\phi(x)]$; $MMD(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}}$.

Apply (1) we have with $f = \mu_P - \mu_Q$: $MMD(P, Q) = \sup_{\|g\|_{\mathcal{H}} \leq 1} \langle g, \mu_P - \mu_Q \rangle$

$= \sup_{\|g\|_{\mathcal{H}} \leq 1} (\mathbb{E}_{x \sim P} [g(x)] - \mathbb{E}_{x \sim Q} [g(x)])$, rename g to f .

$$MMD(P, Q) = \sup_{\|f\|_{\mathcal{H}} \leq 1} \mathbb{E}_{x \sim P} [f(x)] - \mathbb{E}_{x \sim Q} [f(x)] \quad \text{QED}$$

Problem 6.

1. For finite Ω the supremum is attained by $E^* = \{x \in \Omega : P_D(x) \geq P_{D'}(x)\}$.

$$TV(D, D') = 2 \sum_{x \in E^*} (P_D(x) - P_{D'}(x)) = \sum_{x \in \Omega} |P_D(x) - P_{D'}(x)| \quad \text{QED}$$

2. Yes. Put $\mathcal{B} = \{I(h) \subseteq \Omega \mid h \in \mathcal{H}\}$. the sigma algebra generated by hypo set.

$$d_{\mathcal{H}}(D, D') = 2 \sup_{E \in \mathcal{B}} |P_D(E) - P_{D'}(E)| = TV_{\mathcal{B}}(D, D') \quad \text{QED}$$

3. For discrete $d(x, y) \in \{0, 1\}$. Jensen's inequality shows $\frac{1}{2} TV(D, D')$ is

the minimum mass that must be moved to transform D to D' :

$$\mathbb{E}_{\gamma} [d(x, y)] = \gamma(\{(x, y) : x \neq y\}) = 1 - \gamma(\{(x, y) : x = y\}) = 1 - \sum_{x \in \Omega} \gamma(x, x) \quad \text{where } \gamma \text{ is joint PMF-margin } p \text{ and } q$$

Fix any coupling γ . For each x , $\gamma(x, x) \leq \min\{p(x), q(x)\}$. Hence,

$$\sum_x \gamma(x, x) \leq \sum_x \min\{p(x), q(x)\}, \quad \mathbb{E}_{\gamma} [d(x, y)] \geq 1 - \sum_x \min\{p(x), q(x)\} \quad (1)$$

Define $\gamma^*(x, x) = \min\{p(x), q(x)\}$ for all $x \in \Omega$. allocate the leftover mass off-diag

$$p^{\text{rem}}(x) = p(x) - \gamma^*(x, x) \geq 0, \quad q^{\text{rem}}(x) = q(x) - \gamma^*(x, x) \geq 0, \quad \sum_x p^{\text{rem}}(x) = \sum_x q^{\text{rem}}(x)$$

$$\gamma^{\text{rem}}(x, y) = \frac{p^{\text{rem}}(x) q^{\text{rem}}(y)}{\sum_z q^{\text{rem}}(z)} \quad \text{set } \gamma = \gamma^{\text{diag}} + \gamma^{\text{rem}}, \quad x \in \Pi(p, q)$$

$$\sum_x \gamma^*(x, x) = \sum_x \min(p(x), q(x)). \quad \text{inequality (1) is tight for } \gamma^*.$$

$$\begin{aligned} \min_{\gamma} \mathbb{E}_{\gamma} d &= 1 - \sum_x \min\{p(x), q(x)\} = \frac{1}{2} \sum_x (p(x) + q(x) - 2 \min\{p(x), q(x)\}) \\ &= \frac{1}{2} \sum_x |p(x) - q(x)| = \frac{1}{2} TV(D, D') \quad \text{QED} \end{aligned}$$

In general by Jensen's (linearity of $\mathbb{E}$ and convexity of the indicator) matching as much mass on the diagonal as possible is optimal

Problem 7.

1. Sampling $z \sim q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \Sigma_\phi(x))$ is written as.

$z = \mu_\phi(x) + \Sigma_\phi(x)^{\frac{1}{2}} \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$; making the stochastic node independent of parameters so the ELBO gradient $\nabla_\phi \mathbb{E}_q[\cdot]$ can be taken inside the $\mathbb{E}$ and estimated by Monte Carlo with low variance.

2. An ℓ_2 recon term corresponds to Gaussian likelihood.

Replacing it with $\|x - \hat{x}\|_2$ yields the negative likelihood of a Laplace distribution, i.e. choose $p_\theta(x|z) = \text{Laplace}(\hat{x}_\theta(z), b) \Rightarrow -\log p_\theta(x|z) \propto \|x - \hat{x}\|_2$. With that likelihood the ELBO (KL + data term) again equals the total loss so the variational-Bayes interpretation survives.

3. Let $X = [x^{(1)}, \dots, x^{(n)}]$, $\|x - PQX\|_F^2 = \text{tr}((I - PQ)XX^T(I - PQ)^T)$.

is minimised when PQ is the projection A onto the top- d principal component subspace of X . At optimum: $PQ = A = A^T$ (idempotent symmetric)

choose $P = U_d^T$ and $Q = U_d$ where the columns of U_d are the first d left singular vectors of X .

Because $A^2 = A$ and $A^T = A$, it is orthogonal projector. QED.